

```
Explorer
SOA
  api_session_01
    .venv
    Include
    Lib
    Scripts
    pyvenv.cfg
    app.py
    requirements.txt
    bai1.docx
    Bai2.docx
    Bai3.docx
    Bai4.docx
    Bai5.docx
    SOA - B1_ Giới t...

api_session_01 > app.py > ...
1 from flask import Flask, jsonify, request
2 app = Flask(__name__)
3 _next = 1
4 BOOKS = [{"id": 1, "title": "Clean Code", "author": "R. Martin"}]
5 def find(bid):
6     return next((b for b in BOOKS if b["id"] == bid), None)
7 @app.route("/books", methods=["GET"])
8 def list_books():
9     n = int(request.args.get("limit", 100))
10    return jsonify(BOOKS[:n]), 200
11 @app.route("/books/<int:bid>", methods=["GET"])
12 def get_book(bid):
13     book = find(bid)
14     if not book: return {"error": "not found"}, 404
15     return jsonify(book), 200
16 @app.route("/books", methods=["POST"])
17 def create_book():
18     global _next
19     body = request.get_json(silent=True) or {}
20     t, a = body.get("title"), body.get("author")
21     if not t or not a:
22         return {"error": "need title+author"}, 400
23     book = {"id": _next, "title": t, "author": a}
24     _next += 1; BOOKS.append(book)
25     return jsonify(book), 201, {"Location": f"/books/{book['id']}"}
26 @app.route("/books/<int:bid>", methods=["PUT", "DELETE"])

def modify_book(bid):
    book = find(bid)
    if not book: return {"error": "not found"}, 404
    if request.method == "PUT":
        book.update(request.get_json(silent=True) or {})
        return jsonify(book), 200
    BOOKS.remove(book)
    return "", 204
if __name__ == "__main__":
    app.run(host="127.0.0.1", port=5000, debug=True)

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA/api_session_01
$ curl http://127.0.0.1:5000/books
[
  {
    "author": "R. Martin",
    "id": 1,
    "title": "Clean Code"
  }
]

THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA/api_session_01
$ curl http://127.0.0.1:5000/books/1
{
  "author": "R. Martin",
  "id": 1,
  "title": "Clean Code"
}
```

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/soa/api_session_01
$ curl -i -X POST localhost:5000/books \
-H "Content-Type: application/json" \
-d '{"title": "DDIA", "author": "Kleppmann"}'
HTTP/1.1 201 CREATED
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 11 Sep 2026 13:56:58 GMT
Content-Type: application/json
Content-Length: 58
Location: /books/1
Connection: close

{
  "author": "Kleppmann",
  "id": 1,
  "title": "DDIA"
}
```

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/soa/api_session_01
$ curl -i -X PUT localhost:5000/books/1 \
-H "Content-Type: application/json" \
-d '{"title": "CC 2nd ed."}'
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 11 Sep 2026 13:57:20 GMT
Content-Type: application/json
Content-Length: 64
Connection: close

{
  "author": "R. Martin",
  "id": 1,
  "title": "CC 2nd ed."
}
```

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/soa/api_session_01
$ curl -i -X DELETE localhost:5000/books/2
HTTP/1.1 404 NOT FOUND
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 11 Sep 2026 13:57:47 GMT
Content-Type: application/json
Content-Length: 27
Connection: close

{
  "error": "not found"
}
```